GLOBAL AI HACKATHON 2026 · TEAM STACKOVERFLOW

BestBefore

A Bifrost plugin that checks every AI answer against today's web, and keeps stale answers out of your semantic cache.

Statement 04 · Open-Source Bifrost Gateway + Statement 06 · Stale-Answer Detector (Tavily)

github.com/AlgoMasterArpit/HackCulture

Apache 2.0

THE PROBLEM

Models answer from the past. Caches replay it.

USER, 11 NOV 2026

What is the latest stable version of Python?

GEMINI 2.5 FLASH

The latest stable version of Python is 3.12.

WIKIPEDIA, FOUND BY TAVILY

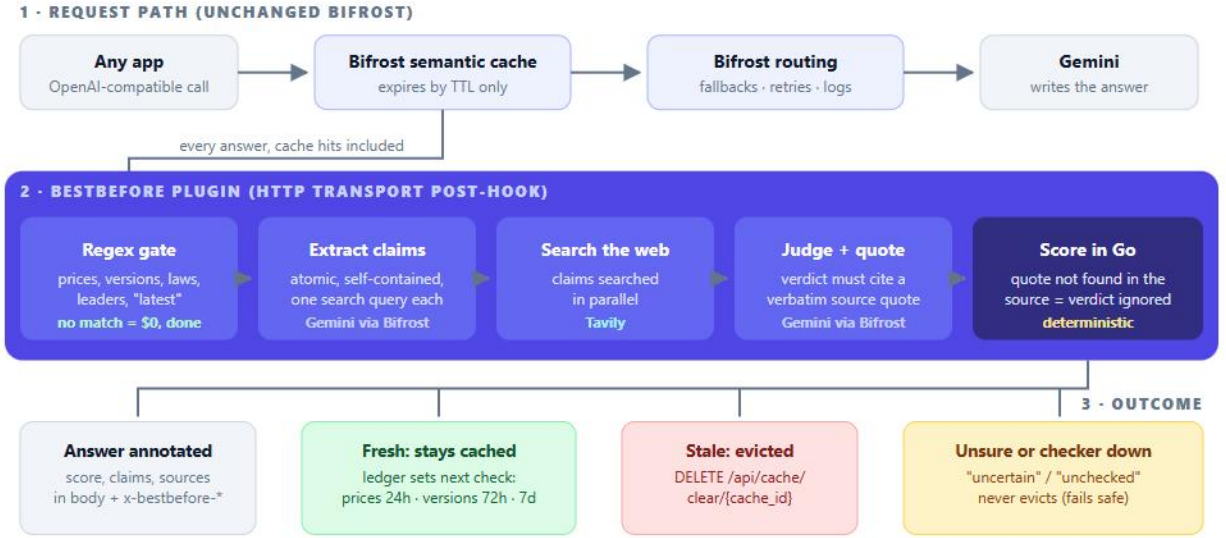
"As of October 2026(Python 3.15 is the latest stable release." (verbatim, page markup included)

A gateway's semantic cache then serves that answer to **every** user who asks again, until the TTL runs out.

In Bifrost today, the clock is the only thing that removes a cached answer: `isExpiredEntry` checks `expires_at` and nothing else (`plugins/semanticcache/search.go:252`), and expired entries are only cleaned up lazily (issue #2414).

WHAT WE BUILT

One plugin. Every answer checked. No app changes.



WHY YOU CAN TRUST A "STALE" FLAG

The model gives advice. Go makes the decision.

The judge must prove it

Every verdict needs a word-for-word quote from the cited source. A contradiction must name a new value whose words are in that quote and not in the claim. Otherwise the verdict is thrown away, with the reason recorded.

Fails safe

Model down, search down or unsure: the answer is marked `unchecked` or `uncertain` and is never evicted. An outage of the checker cannot wipe your cache.

Free when nothing can change

A regex gate skips answers with no prices, versions, laws, leaders or "latest". Zero tokens, zero searches.

Broken on purpose

A separate AI reviewer was told to break it. It found 12 problems, 3 high-severity. The 3 high-severity ones are fixed, and each fix has a test that fails without it.

RESULTS · 60 HAND-LABELLED ANSWERS, LIVE GEMINI + TAVILY

Catches stale answers without crying wolf.

80%

of stale answers flagged stale

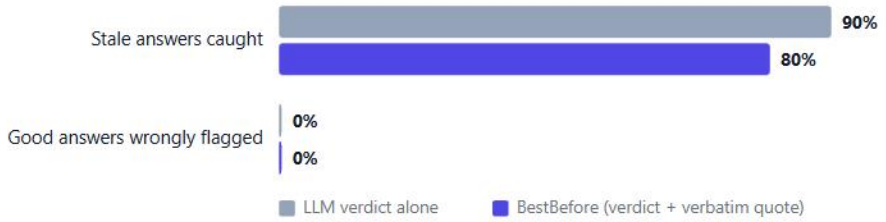
0%

of good answers wrongly flagged stale; **100%** of stale flags were right

85%

of timeless answers checked at zero cost

What the quote check costs on clean answers (its job is attacks, see docs/REVIEW.md)



From eval/run.py. Rule tuned on this set after a first run that caught **65%**; both runs published.

RESULTS · LIVE SEMANTIC CACHE

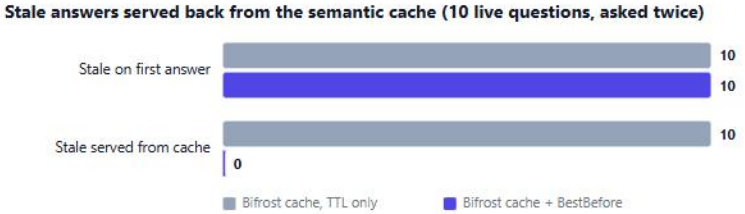
A TTL cache replays stale answers. BestBefore stops it.

10

stale answers served from a TTL-only Bifrost cache

0

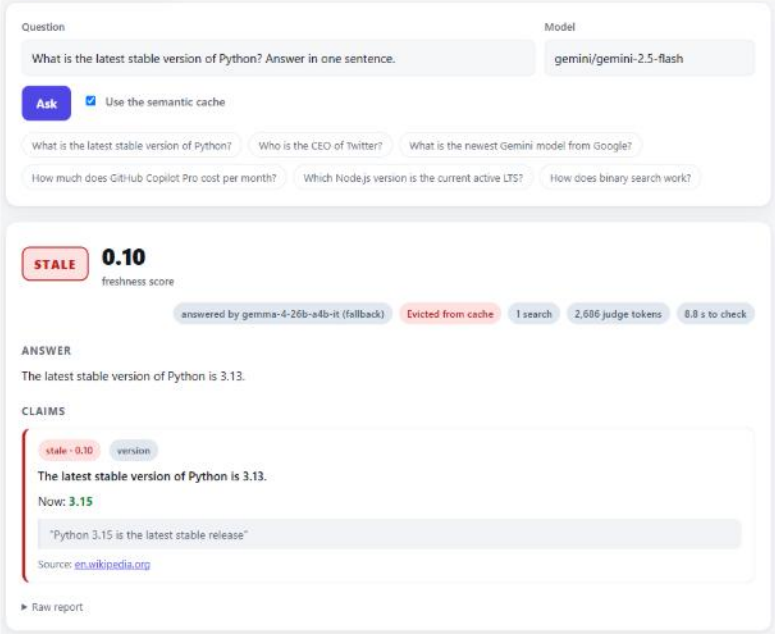
with BestBefore



eval/cache.py · the same time-sensitive questions asked twice through each gateway

WORKING PROTOTYPE

Ask through the gateway, or paste any answer.



HOW EACH TECHNOLOGY CONTRIBUTES

Take either away and nothing works.

TECHNOLOGY	WHAT IT DOES IN BESTBEFORE
Bifrost plugin hooks	The HTTP transport post-hook sees every answer, cache hits included, and rewrites body and headers. The pre-hook serves the `bestbefore/verify` model.
Bifrost semantic cache	`cache_debug.cache_id` says which entry produced the answer; `DELETE /api/cache/clear/{id}` evicts it.
Bifrost routing	The judge's own calls go back through the gateway: fallbacks, retries on 503s, logs and budgets for free.
Tavily search	Live evidence per claim: URL, page text, date and a relevance score that weights the freshness score.
Gemini	Answers the questions, then extracts claims and judges them against the evidence.

HONEST LIMITS · WHAT'S NEXT

What it does not do yet.

- Streaming responses pass through unchecked.
- Checking adds latency to answers with time-sensitive claims (p95 **15.7s**).
- One grounded source is enough, so a highly ranked outdated page can still pass.
- The recheck ledger is in memory, per gateway process.
- Upstream PR drafted in Bifrost's template (docs/UPSTREAM_PR.md), building on proposal #9748.
- Check streams after they finish, and evict in the background.
- Store recheck times in the vector store so replicas share them.
- Require two sources on different hosts before evicting.

RUN IT

Three commands.

```
git clone https://github.com/AlgoMasterArpit/HackCulture.git
cp .env.example .env     # add Gemini + Tavily keys
docker compose up -d --build
```

Team stackoverflow · github.com/AlgoMasterArpit/HackCulture